



Medical Microbiology

Third stage- College of Dentistry

Lab.7

Streptococcus

Prepared by

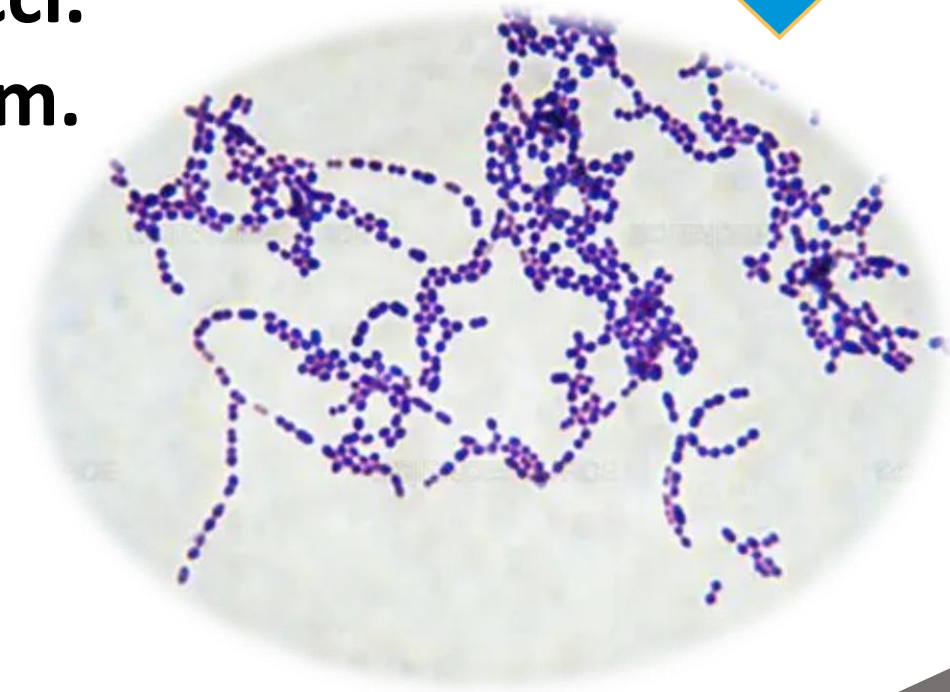
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Learning Objectives:

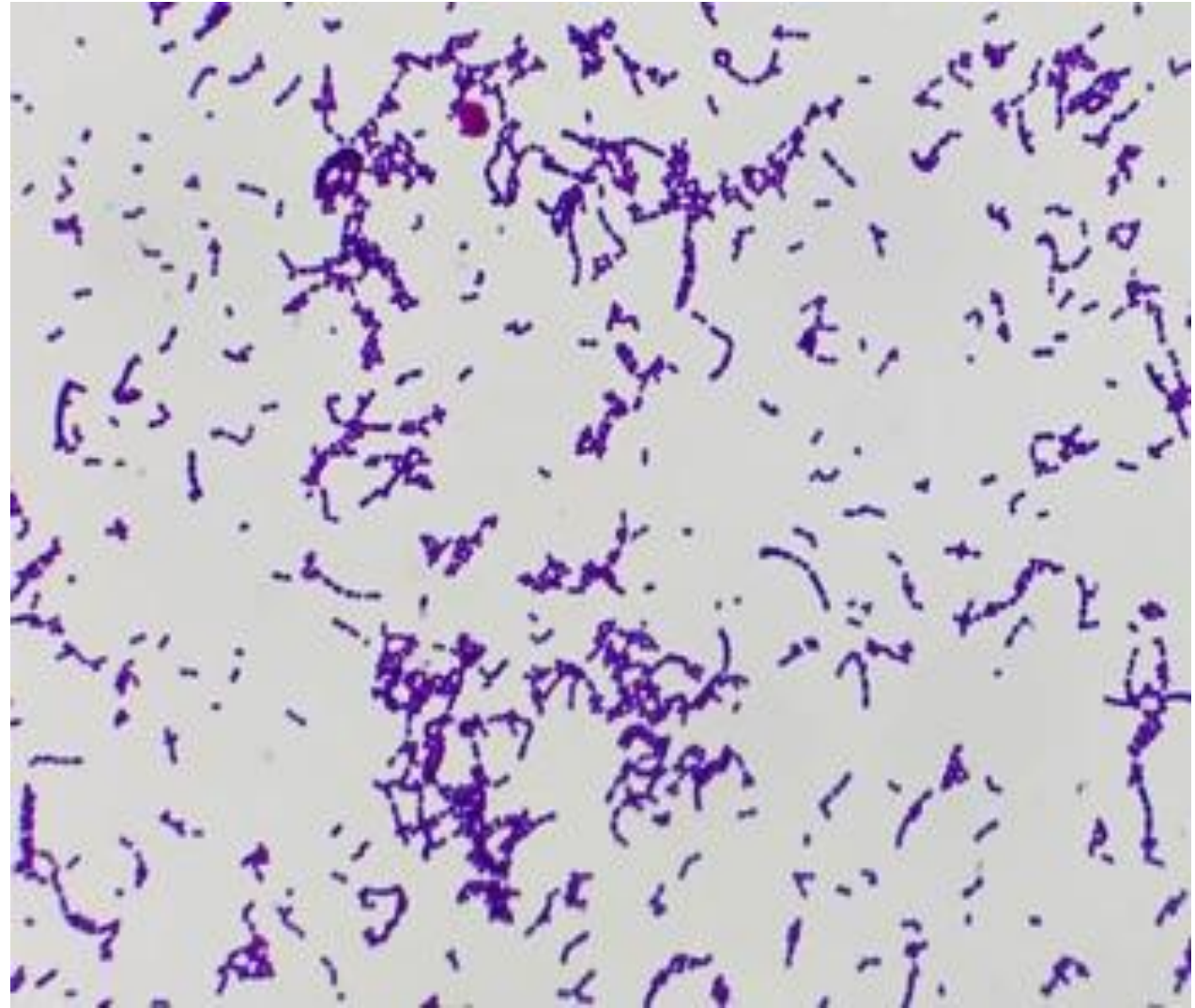
After this lab. You must be able to:

- 1-Describe streptococci under microscope.
- 2-Distinguish between types of streptococci.
- 3-Described Lancefield classification system.



Morphology of streptococci

- ✓ Gram positive
- ✓ Cocci in long chains
- ✓ Facultative anaerobic
- ✓ Non- motile
- ✓ Non-spore forming
- ✓ Capsule – some strains
- ✓ Catalase negative



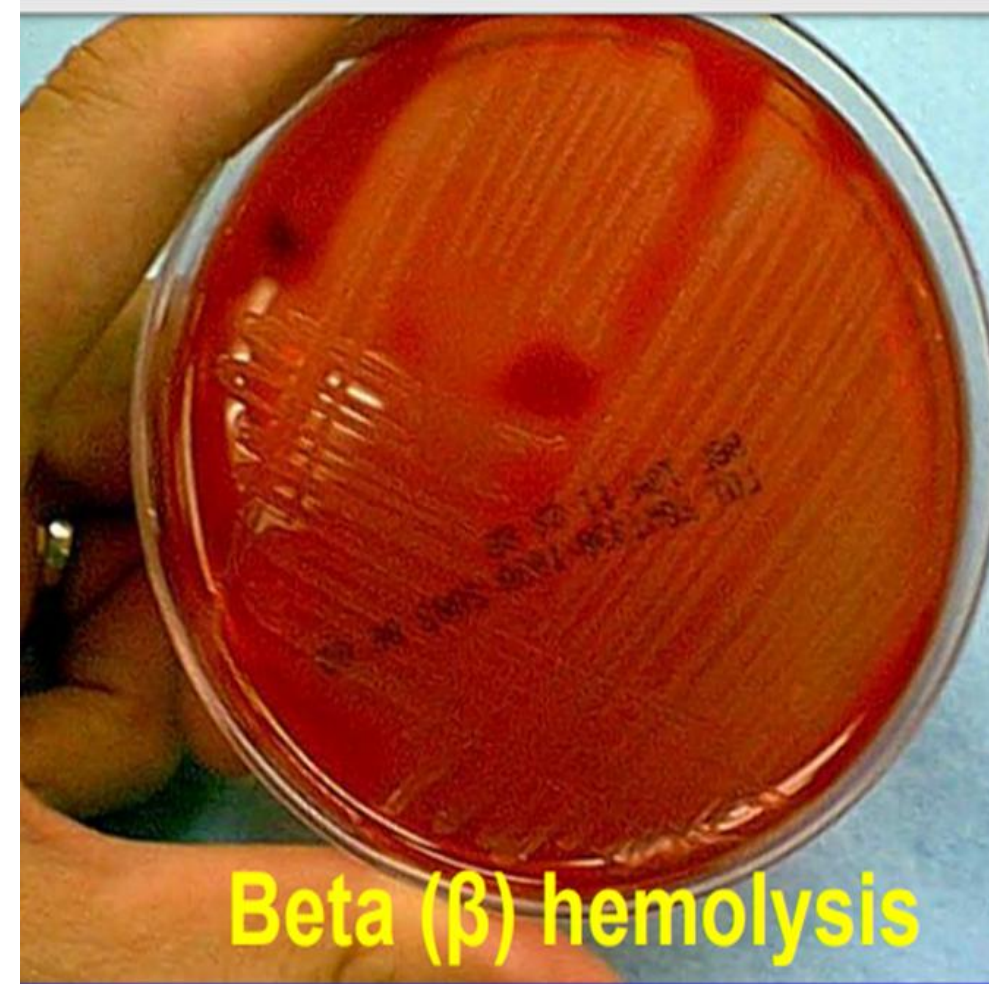
Classification based on:-

1-Hemolysin on blood agar

A) Complete hemolysis (Beta):

Clear zone surrounding colonies, Ex:

**Streptococcus
pyogenes**



Beta (β) hemolysis

**Complete hemolysis
with clearing around colony**

1- Hemolysin on blood agar

B) Partial hemolysis (Alpha):

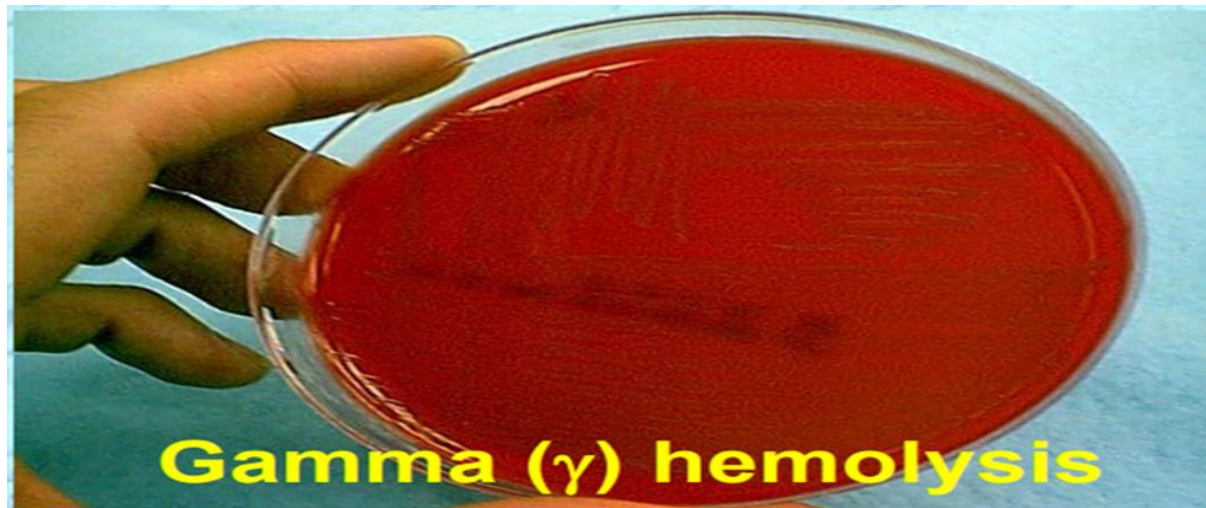
Greenish zone surrounding colonies, Ex:

**Streptococcus
pneumoniae, agalactiae**

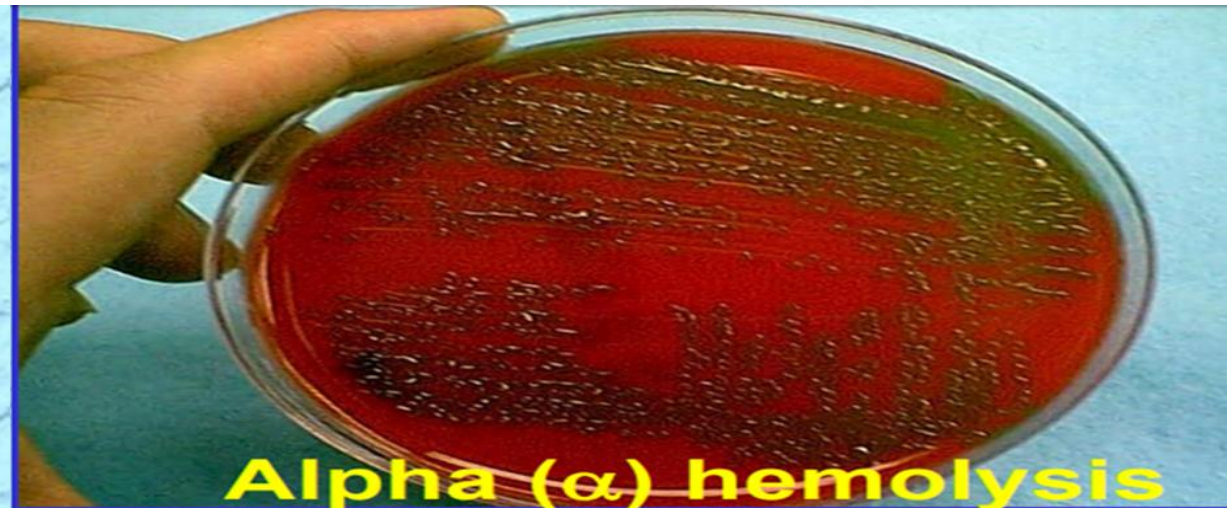
C) Non hemolysis (Gamma) :

non hemolytic non greenish zone surrounding colonies,

Ex: Streptococcus bovis

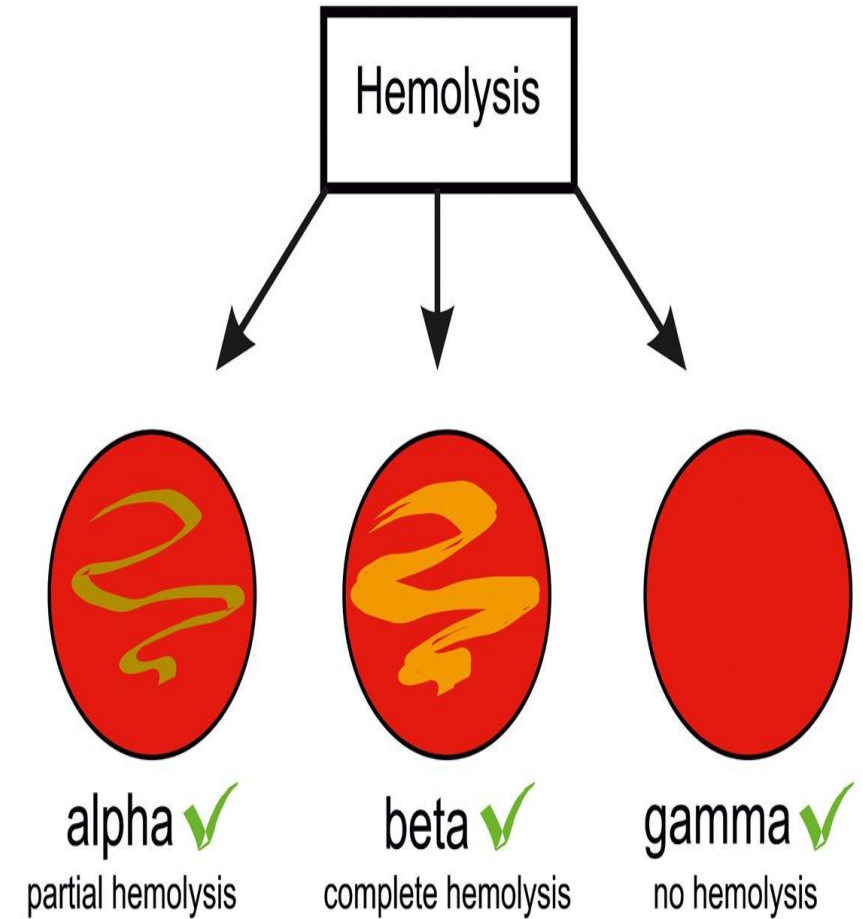


Non-hemolytic



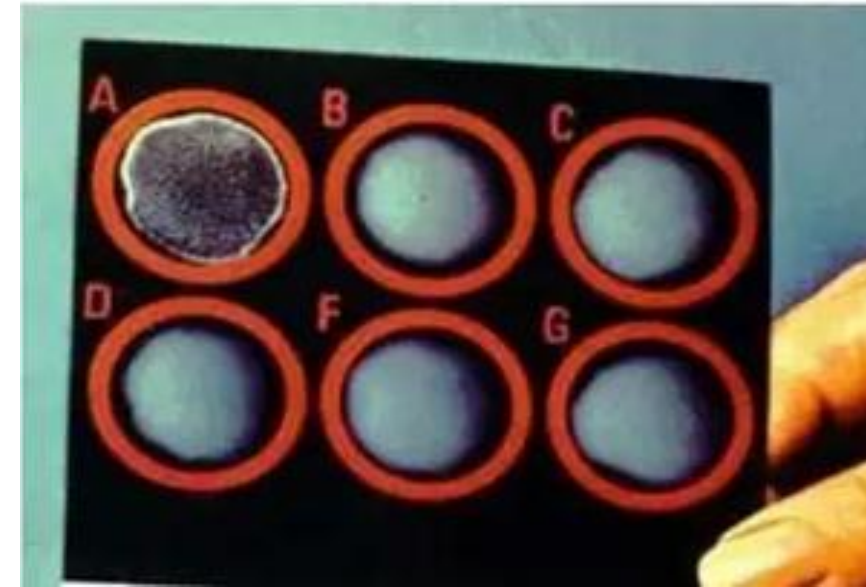
**Partial hemolysis
Green discoloration of media**

Hemolysis Sheep Blood Agar



2-Lancefield classification system

- The Lancefield classification system categorizes beta-hemolytic streptococci into distinct serological groups based on differences in their cell wall carbohydrate antigens.
- These antigenic groups, which help identify pathogenic species, are designated by the letters A through T.
- Lancefield serological groups A,B,C,D,F and G are the ones that normally infect humans.



Important Lancefield Groups

- Group A are mostly human, primate pathogens.
- Groups B and C contain most veterinary pathogens.

Virulence Factors of Importance Cellular

- Capsule(Adhesin)
 - Resists phagocytosis
- Fimbriae
 - Attachment and colonization
- Streptokinase
 - Activates complement
 - Lyses fibrin
- Hemolysins: lyses of RBC.

Differential Laboratory Tests:

Specific biochemical and sensitivity tests are used to confirm the species after the hemolysis pattern is determined.

1-Bacitracin Sensitivity Test:

Purpose: To differentiate Group A *S. pyogenes* from other beta-hemolytic streptococci.

Sensitive: *S. pyogenes*.

Resistant: *S. agalactiae* and most others.



2-CAMP Test:

The CAMP test is used to identify *Streptococcus agalactiae* (Group B *Streptococcus*).

Principle:

S. agalactiae produces a CAMP factor that enhances the hemolysis caused by *Staphylococcus aureus* β -hemolysin.

Procedure:

- 1-Streak *S. aureus* in a straight line on a blood agar plate.
- 2-Streak the test organism (suspected *Streptococcus*) perpendicular to it, without touching.
- 3-Incubate at 37°C for 18–24 hours.

Result:

Positive: Arrowhead-shaped hemolysis at the junction
→ *S. agalactiae*.

Negative: No enhanced hemolysis → other *Streptococcus* species.



3-Optochin Sensitivity Test:

Purpose: To differentiate *S. pneumoniae* (alpha-hemolytic) from the Viridans group (alpha-hemolytic).

Resistant: *S. Viridans*.

Sensitive: *S. pneumoniae*.

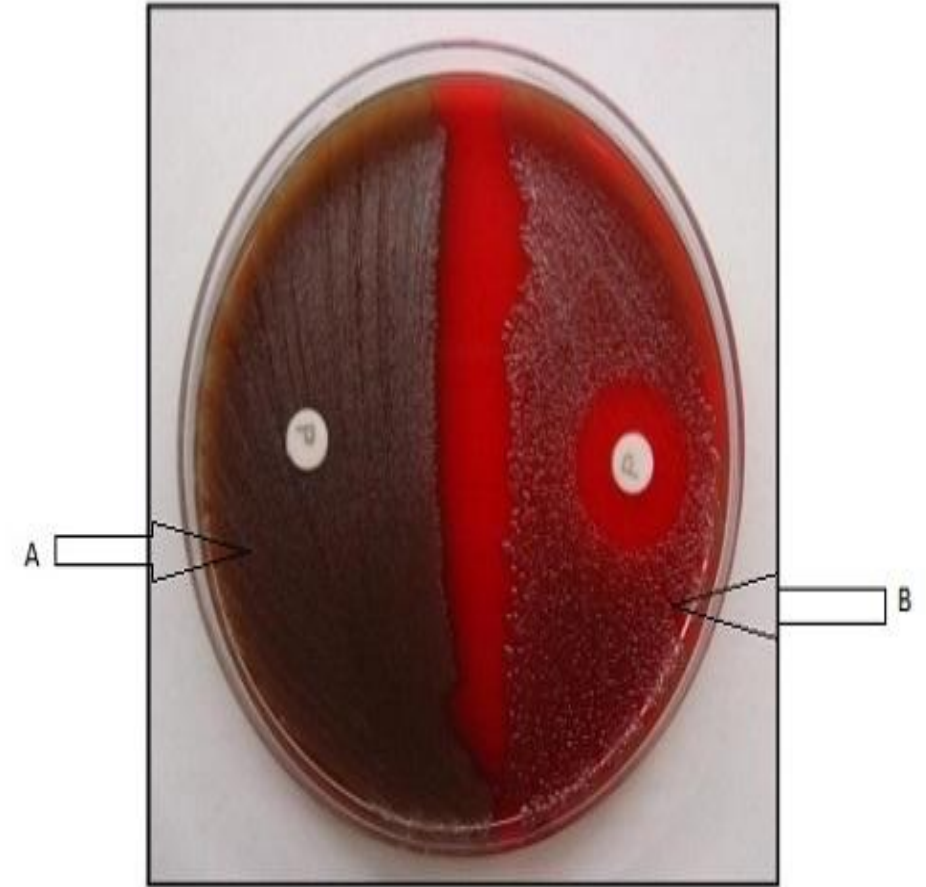


Fig. Optochin Test using optochin disk

A. *S. viridians*, Resistance to optochin

B. *S. pneumoniae*, Susceptible to optochin

4-Bile Esculin Hydrolysis Test (BEA):

Purpose: To identify *Group D Streptococci* and *Enterococci*.

Principle: These bacteria can grow in the presence of bile and hydrolyze esculin into esculetin, which reacts with ferric ions to produce a **black color**.

Result:

Positive: Medium turns **black** → Like: *Enterococcus faecalis*.

Negative: No color change (medium remains brown).





Any Question?



Thank you